WANG 500/520 SERIES ELECTRONIC CALCULATORS

DESCRIPTION

The Wang 500/520 brings to the engineering and scientific user a moderately priced advanced programmable calculator at an excellent cost/performance ratio. Great emphasis on human engineering provides the user with a unit designed for both manual and programmed use.

Scientific and engineering calculations are the main features of the unit. Such functions as X^2, $\sqrt{X}$, $\mathrm{Log}_e X$, $\mathrm{Log}_{10} X$, e^x, 10^x, 1/X, |X|, SIN X, COS X, TAN X, SIN^{-1} X, COS^{-1} X, TAN^{-1} X and INT X are all fast and accurate single-keystroke applications. Yet another feature of great importance is the availability of 16 individual calculators to perform single keystroke arithmetic operations, such as +, −, ×, ÷, TOTAL, STORE, and RECALL for ease of data manipulation. The unit can display ten digits with a sign and decimal point; and a two-digit exponent with a sign.

Optional magnetic tape cassette and printer are available for flexibility in calculating applications.

DISPLAY

Type: ½-inch full character display tubes.

Capacity: 10-digits, sign and separate decimal-point tube, two-digit exponent and sign.

Dynamic Range: $\pm 10^{-99}$ to $\pm 10^{+99}$.

Decimal Settings: Display – automatic floating decimal point and scientific notation (the floating decimal point is automatically converted to scientific notation from $>\pm 10^{+10}$ or $<\pm 10^{-10}$).

Decimal Settings: Printer – floating decimal point, fixed decimal point from 0 to 9 decimal places and scientific notation.

KEYBOARD

Keyboard: Long-life microswitches with color-coded key tops.

Data Entry: Standard ten-key keyboard plus decimal. Exponent can be entered separately. All entries are displayed as entered.

Arithmetic: +, −, ×, ÷, X^2, $\sqrt{X}$, 1/X, Exchange

Logarithmic: $\mathrm{LOG}_e X$, LOG_{10} X

Exponential: e^x, 10^x

Miscellaneous: |X|, Integer of X

Trigonometric Functions: (A switch is provided to permit input of either degrees or radians to the trigonometric keys.) SIN(X), COS(X), TAN(X), SIN^{-1}(X), COS^{-1}(X), TAN^{-1}(X). To facilitate special conversions, a DEGREES-TO-RADIANS and RADIANS-TO-DEGREES key is provided.

Special Two-Step Commands: π (Pi), (pause ½ second), decimal shift, 0 to 15 places, right or left.

Other Functions: The availability of 32 user defined subroutines.

SPEEDS	
Function	Approximate Speed
Addition/Subtraction	1.7 ms
Multiplication/Division	12.0ms, 17.6ms
X^2, 1/X	12.0ms, 17.6ms
10^x, Log_{10}X	44.4ms, 35.3ms
$\sqrt{X}$	84.4ms
e^x, Log_eX	61.8ms, 47.2ms
Trigonometric Functions	82ms to 238ms
Memory Read/Write cycle time	5 μs

500 SERIES MEMORY CONFIGURATIONS

MODEL	BITS	REGISTERS	STEPS	TOTAL* REGISTERS
500-0	2048	16	64	24
500-1	3076	16	192	40
500-2	4096	16	320	56
520-2	4096	16	312	55
520-6	8192	16	824	119
500-14	16384	16	1848	247

***Program steps can be converted to storage registers. It takes 8 steps to make up one storage register.**

WANG LABORATORIES, INC.
836 NORTH STREET, TEWKSBURY, MASSACHUSETTS 01876, TEL. (617) 851-7311, TWX 710 343-6769, TELEX 94-7421

ADDRESSING STORAGE REGISTERS

Direct Addressing 500/520: Single-step addressing of each of the basic sixteen registers is provided. Each register is capable of 8 operations: STORE, RECALL, +, –, X, ÷, TOTAL and interchange (with the display register). The arithmetic functions are performed without recall or re-entry of data, and the answer is retained in the register. Direct addressing in extra storage registers provides STORE and RECALL operations only.

Indirect Addressing (520 Series only): All registers in the 520 as well as the registers of extended R.A.M. can be addressed by referencing the contents of any one of the sixteen basic registers. By means of indirect addressing, all registers are capable of performing +, –, X, ÷, STORE, RECALL, TOTAL and INTERCHANGE.

PROGRAMMING

Method: Entered via keyboard into MOS memory by pressing keys in proper sequence. LEARN mode, by loading from a magnetic tape cassette (optional feature) or from the Mark Sense Card Reader (peripheral).

Capacity: 64, 192, 320, 312, 824, or 1848 steps (dependent upon memory size). Total capacity with one cassette is 32,000 steps.

Unconditional Branching: A SEARCH XX statement can be programmed to branch to the corresponding MARK XX in a program. 255 separate marks can be programmed in any location in memory.

Conditional Branching: Jump two steps if condition exists. JUMP IF 0 (display is zero), JUMP IF NOT 0 (display is not zero), JUMP IF + (display is positive).

Subroutines: Five level subroutine capability.

Skip Commands. (520 only)

1. Alpha J = 0 – Skip two steps if register 0 is numerically equal to the window.
2. Alpha J if + – Skip two steps if register 0 is numerically greater than or equal to the window.
3. Alpha sine – Skip two steps if register 0 is numerically less than the window.

COMPARE COMMANDS

1. I/O 0200 – Skip two steps if register 0 is (alpha numerically) equal to the window.
2. I/O 0300 – Skip two steps if register 0 is not (alpha numerically) equal to the window.

Test for Error: JUMP IF ERROR tests for illegal mathematical operation, overflow or non-existing address (see next paragraph).

Error Lights: MACHINE ERROR: Indicates improper loading of tape. PROGRAM ERROR: Indicates illegal operations (see previous paragraph).

Program Debugging: To detect, locate and remove mistakes written into a program or program routine. In the LEARN mode, the program step number and the code associated with that step will be displayed. The following commands are available to the user that will assist him in his debugging procedures: SET PROGRAM COUNTER, VERIFY PROGRAM and STEP.

Commands available only on the 520 are: BACKSTEP, INSERT PROGRAM STEP, DELETE PROGRAM STEP, and SEARCH MARK and STOP.

PROGRAM TAPE UNIT (OPTIONAL)

Record Program: A record program command can record programs from the MOS memory onto a magnetic tape.

Load Program – Manual: Control keys are provided to load up to 1848 steps of program instructions into memory in one operation.

Load Program – Automatic: Loads and executes a block of program steps (max. 32,000) by a command entered into memory. With this command programs may be chained automatically.

Go to a Block: (520 only) With this command it is possible to load a particular block by passing the intervening blocks. The display indicates how many blocks are to be bypassed, thus it is possible to retrieve any programs easily and efficiently from a single tape cassette.

Data Storage: (520 only) On the 520 it is possible to selectively store data on tape, and to selectively recall data from the tape. When storing, the display indicates how many registers will be recorded on the tape. When retrieving data from tape, the display indicates in which storage register the data will be placed.

PRINTER (OPTIONAL)

Printing Method: Parallel drum printed on plain paper rolls. Up to 20 columns can be printed on one line at a rate of 160 lines per minute.

Print Format: MANUAL: print format is stored in the memory from the keyboard. Printing occurs after depression of a single key in either a floating, 2 through 9 decimal places, or exponential form. AUTOMATIC FROM PROGRAM: A two-step command designating format, identifying letter and operation is required in the program. Print format for an individual command can be floating, 0 to 9 decimal places, exponential form, or paper feed. AUTOMATIC PRINT: This feature enables a user to generate a complete hard copy record of all manual operations.

Trace Mode: Alpha Log_{10} or Alpha Log_e turns the TRACE mode on. The printer will print the contents of the window as well as the operation while executing a program. This feature is programmable.
Alpha 10^x, Alpha e^x or PRIME will turn the TRACE mode off.

Identifying Letter: A programmed print command designating one of 16 separate identifying letters is provided.

List Program – Learn Print: Generates a hard-copy output of the program operation, printing the program step number, program instruction code and the operation symbol.

Paper Feed: A key is provided for manual paper feed as well as a two-step program command. Paper feed rate is 400 lines per minute.

SOFTWARE CAPABILITY

An extensive library of programs is available to further enhance the user's programming capabilities.

FIELD RETROFITTING UPDATING A USER'S SYSTEM

Wang's modularity concept allows users to update their systems easily, efficiently and inexpensively on location.

GENERAL

Dimensions: 9 inches high, 20 inches wide, 20½ inches deep.
Weight: Net 39 lbs.
Power: 115 volts ac ±10%, 60Hz, 50 watts (standard); 230 volts ac ±10%, 50Hz, 50 watts (optional, no charge).

WARRANTY

Full warranty of 90 days. Maintenance contracts available.

PERIPHERAL

514 MARK SENSE CARD READER

The Mark Sense Card Reader allows off-line programming for software development without the use of the calculator, thus freeing the keyboard for other calculating applications.

ORDERING SPECIFICATIONS

All Models: Self-contained, desktop electronic programmable calculators. They all include; ten-key keyboard, decimal, scientific notation, change sign, clear display, set exponent. Also with sixteen separate calculators that will perform addition, subtraction, division, and multiplication, along with STORE, RECALL, and TOTAL. Display has ten-digit accuracy with a sign and a two-digit exponent with a sign. Scientific and engineering keys include; $\sqrt{X}$, X^2, |X|, INT X, e^x, 10^x, 1/X, Log_eX, $Log_{10}X$, and trig functions. 32 user-defined subroutine keys. Logic decisions, branching, looping and subroutine capabilities.

MODEL	REGISTER	STEP/STORAGE CAPACITY	TOTAL REGISTERS
500-0	16	64/8	24
500-1	16	192/24	40
500-2	16	320/40	56
520-2	16	312/39	55*
520-6	16	824/103	119*
520-14	16	1848/231	247*

*Models that have indirect addressing capabilities allowing for +, −, ×, ÷, STORE, RECALL, and TOTAL in all registers.

OPTIONS: **All models:**
Magnetic tape cassette – Automatic program loading.
Rotary drum printer – Alpha character labeling, decimal point formatting.

PERIPHERAL: **514 Mark Sense Card Reader** – Off-line programming capability.

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